AGU ECE531 — Computer Vision

Project Research Proposal

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Cross-Modal Attention-Based Infrared-Visible Fusion for Robust Lightweight Object Detection and Semantic Segmentation Under Adverse Conditions

Short Description

This project investigates whether attention-based cross-modal infrared-visible image fusion can systematically improve the performance of lightweight computer vision models (YOLOv8-Nano, U-Net/MobileNetV2) under adverse weather conditions including nighttime, fog, and rain. A dual-branch CNN fusion module with channel and spatial attention mechanisms will be developed, and its impact on downstream detection and segmentation will be evaluated through condition-stratified analysis across multiple RGB-thermal benchmarks.

Problem Definition and Motivation

The deployment of visual perception systems confronts a fundamental reliability paradox. Standard visible cameras are inherently dependent on external illumination and are easily compromised by ambient lighting and adverse weather conditions (Liu et al., 2025). Under scenarios suffering from severe degradation—such as low light, dense fog, or rain—the lack of reliable texture information critically impairs downstream tasks like object detection (Liu et al., 2025). Semantic segmentation accuracy degrades by approximately 60% under rainy conditions (Tremblay et al., 2021). Thermal infrared (IR) sensors detect emitted radiation proportional to object temperature, enabling target detection in darkness and through atmospheric obscurants (Tu et al., 2023). However, IR suffers from low spatial resolution and absence of texture/color information.

The comprehensive IVIF survey by Liu, Wu, et al. (2025) confirms that fusion evaluation is systematically conducted using heavyweight architectures (SegFormer 84M, Swin Transformer 197M) impractical for edge deployment. Whether fusion benefits transfer to sub-5M-parameter models remains unvalidated. Furthermore, the correlation between pixel-level fusion quality metrics (MI, VIF) and actual task-level performance (mAP, mIoU) on lightweight models has never been systematically tested.

Literature Review

The infrared-visible image fusion (IVIF) field has evolved from classical multi-scale transforms through deep autoencoder architectures to task-aware frameworks. DenseFuse introduced the autoencoder paradigm with dense connections and element-wise fusion for IVIF (Li & Wu, 2019). CDDFuse decomposed features into base and detail components through parallel Transformer-CNN branches (Zhao et al., 2023). Task-aware fusion methods incorporated downstream task losses: SeAFusion added segmentation loss to the fusion objective (Tang, Yuan, & Ma, 2022), TarDAL jointly optimized fusion and detection while introducing the M3FD benchmark (Liu, Fan, et al., 2022), and MRFS established mutually reinforcing fusion and segmentation (Zhang et al., 2024).

For attention mechanisms, SE-Net established channel recalibration through squeeze-and-excitation blocks (Hu et al., 2018), CBAM added spatial attention via pooling statistics (Woo et al., 2018), and CMX demonstrated cross-modal transformer fusion for RGB-X segmentation (Zhang, Liu, et al., 2023). On lightweight models, YOLOv8-Nano achieves competitive detection with 3.2M parameters (Ultralytics, 2023), while U-Net with

MobileNetV2 encoder provides efficient segmentation at approximately 3.4M parameters (Sandler et al., 2018; Ronneberger et al., 2015).

Baseline Methods

Baseline 1 — DenseFuse (Li & Wu, 2019): Deep autoencoder fusion with pixel-level reconstruction. IEEE TIP publication with over 3,000 citations. Pre-trained model available on GitHub. Represents the dominant deep learning pixel-reconstruction paradigm.

Baseline 2 — Laplacian Pyramid Fusion (Burt & Adelson, 1983): Classical multi-scale decomposition with max-absolute fusion rule. Zero trainable parameters, deterministic, content-agnostic. Establishes the classical fusion floor.

Additional baselines: RGB-Only and IR-Only single-modality inputs (no fusion) to quantify the "adverse condition gap."

Research Gap

Despite rich IVIF literature, no existing study provides: (1) systematic evaluation of cross-modal attention-based fusion specifically targeting sub-5M-parameter downstream models; (2) condition-stratified analysis (day vs. night vs. fog vs. rain) quantifying under which scenarios fusion provides maximal benefit for lightweight detectors; (3) correlation analysis between traditional pixel-level fusion quality metrics and actual task-level performance on lightweight models.

Novel/Improved Method and Steps

The proposed architecture employs a three-stage modular pipeline:

Stage 1 — Dual-Branch Feature Extraction: Two separate CNN encoders process RGB and IR through independent pathways (4-stage Conv-BN-ReLU blocks, 3→64→128→256→512 channels, stride-2 downsampling). Each branch ~2.3M parameters.

Stage 2 — Cross-Modal Attention Fusion: Channel attention (SE-Net-inspired MLP, reduction ratio r=16) learns informative channels. Spatial gating (pixel-wise sigmoid) determines RGB vs. IR dominance per location. Fused output: F_ctx = G ⊙ F'_RGB + (1−G) ⊙ F'_IR.

Stage 3 — Feature Adapter + Downstream Evaluation: Lightweight decoder (512→3 channels, ~0.3M params) converts fused features to 3-channel images for unmodified YOLOv8-Nano and U-Net.

Block Diagram: I_RGB → CNN Encoder (Branch 1) → [Channel Attention + Spatial Gating] ← CNN Encoder (Branch 2) ← I_IR → F_ctx → Feature Adapter → I_adapted (3-ch) → YOLOv8-Nano / U-Net → Detection / Segmentation Results

Dataset

Dataset	Description	Role
M3FD	4,200 aligned RGB-IR pairs, 1024×768, day/night/fog/rain labels, 6-class YOLO-format annotations. Source: github.com/JinyuanLiu-CV/TarDAL (Liu, Fan, et al., 2022)	Primary: training + condition-stratified detection evaluation
MSRS	1,444 paired RGB-IR, 480×640, 9-class segmentation annotations, day/night splits. Source: github.com/Linfeng-Tang/MSRS (Tang et al., 2022)	Primary: training + segmentation evaluation

RoadScene	221 IR-visible pairs, diverse urban driving. Source: github.com/hanna-xu/RoadScene (Xu et al., 2020)	Extended: cross-dataset generalization + fusion quality
LLVIP	16,836 pairs, low-light pedestrian detection. Source: bupt-ai-cz.github.io/LLVIP (Jia et al., 2021)	Extended: low-light evaluation

Performance Metrics

Detection: mAP@0.5 and mAP@[0.5:0.95] (COCO protocol).

Segmentation: mIoU across all classes; per-class IoU for Person and Car.

Fusion quality: Mutual Information (MI), Entropy (EN), Spatial Frequency (SF), Visual Information Fidelity (VIF), Average Gradient (AG).

Efficiency: Inference FPS, peak GPU memory (GB), total parameter count.

Correlation: Pearson/Spearman between fusion quality and task metrics across all methods and conditions.

Experimental Setup and Hyperparameters

Platform: Google Colab free tier (NVIDIA T4, 15 GB VRAM). PyTorch 2.1+, Ultralytics (YOLOv8), segmentation_models_pytorch (U-Net).

YOLOv8-Nano: COCO pretrained. SGD optimizer, lr = 0.01, momentum = 0.937, weight_decay = 5×10^{-4}, batch size = 16, epochs = 100, input = 640×640.

U-Net/MobileNetV2: ImageNet pretrained encoder. Adam optimizer, lr = 10^{-4}, batch size = 4, epochs = 150, input = 480×640, loss = CrossEntropy + Dice.

Fusion module: Adam optimizer, lr = 2×10^{-4}, batch size = 8, epochs = 100. Training loss: L1 + SSIM + Sobel gradient-aware edge preservation.

Hyperparameters to test: Channel attention reduction ratio $r \in \{8, 16, 32\}$; gradient loss weight $\gamma \in \{1.0, 5.0, 10.0\}$; adapter depth {2-layer, 3-layer}.

Ablation: (A0) full model, (A1) no channel attention, (A2) no spatial gating, (A3) plain concatenation, (A4) no gradient loss. Each trained/evaluated identically.

Paper Draft Outline

Section	Content
Abstract	150–250 words: problem, method, datasets, key quantitative result.
1. Introduction	Adverse condition challenge. RGB/IR complementarity. Research gap: no systematic lightweight evaluation. Contributions.
2. Related Work	Classical fusion (LP, DWT). Deep fusion (DenseFuse, CDDFuse, SeAFusion, TarDAL). Attention (SE-Net, CBAM, CMX). Lightweight models.
3. Proposed Method	Dual-branch encoder. Channel attention + spatial gating. Feature adapter. Training protocol.
4. Experimental Setup	Datasets with URLs. Baselines. Downstream model configs. Metrics. Ablation design.
5. Results & Discussion	Quantitative tables. Condition-stratified analysis. Metric correlation. Gating visualizations. Qualitative overlays.
6. Conclusion	Findings. Limitations (requires aligned pairs). Future: latent prediction, temporal extension.

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